

2.3 Power series

The prime example of a power series is the complex **exponential** function, which is defined for $z \in \mathbb{C}$ by

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

When z is real, this definition coincides with the usual exponential function, and in fact, the series above converges absolutely for every $z \in \mathbb{C}$. To see this, note that

$$\left| \frac{z^n}{n!} \right| = \frac{|z|^n}{n!},$$

so $|e^z|$ can be compared to the series $\sum |z|^n/n! = e^{|z|} < \infty$. In fact, this estimate shows that the series defining e^z is uniformly convergent in every disc in $\mathbb{C}$.

In this section we will prove that e^z is holomorphic in all of $\mathbb{C}$ (it is entire), and that its derivative can be found by differentiating the series term by term. Hence

$$(e^z)' = \sum_{n=0}^{\infty} n \frac{z^{n-1}}{n!} = \sum_{m=0}^{\infty} \frac{z^m}{m!} = e^z,$$

and therefore e^z is its own derivative.

In contrast, the geometric series

$$\sum_{n=0}^{\infty} z^n$$

converges absolutely only in the disc $|z| < 1$, and its sum there is the function $1/(1-z)$, which is holomorphic in the open set $\mathbb{C} - \{1\}$. This identity is proved exactly as when z is real: we first observe

$$\sum_{n=0}^N z^n = \frac{1 - z^{N+1}}{1 - z},$$

and then note that if $|z| < 1$ we must have $\lim_{N \rightarrow \infty} z^{N+1} = 0$.

In general, a **power series** is an expansion of the form

$$(5) \quad \sum_{n=0}^{\infty} a_n z^n,$$

where $a_n \in \mathbb{C}$. To test for absolute convergence of this series, we must investigate

$$\sum_{n=0}^{\infty} |a_n| |z|^n,$$

and we observe that if the series (5) converges absolutely for some z_0 , then it will also converge for all z in the disc $|z| \leq |z_0|$. We now prove that there always exists an open disc (possibly empty) on which the power series converges absolutely.

Theorem 2.5 *Given a power series $\sum_{n=0}^{\infty} a_n z^n$, there exists $0 \leq R \leq \infty$ such that:*

- (i) *If $|z| < R$ the series converges absolutely.*
- (ii) *If $|z| > R$ the series diverges.*

Moreover, if we use the convention that $1/0 = \infty$ and $1/\infty = 0$, then R is given by Hadamard's formula

$$1/R = \limsup |a_n|^{1/n}.$$

The number R is called the **radius of convergence** of the power series, and the region $|z| < R$ the **disc of convergence**. In particular, we have $R = \infty$ in the case of the exponential function, and $R = 1$ for the geometric series.

Proof. Let $L = 1/R$ where R is defined by the formula in the statement of the theorem, and suppose that $L \neq 0, \infty$. (These two easy cases are left as an exercise.) If $|z| < R$, choose $\epsilon > 0$ so small that

$$(L + \epsilon)|z| = r < 1.$$

By the definition L , we have $|a_n|^{1/n} \leq L + \epsilon$ for all large n , therefore

$$|a_n| |z|^n \leq \{(L + \epsilon)|z|\}^n = r^n.$$

Comparison with the geometric series $\sum r^n$ shows that $\sum a_n z^n$ converges.

If $|z| > R$, then a similar argument proves that there exists a sequence of terms in the series whose absolute value goes to infinity, hence the series diverges.

Remark. On the boundary of the disc of convergence, $|z| = R$, the situation is more delicate as one can have either convergence or divergence. (See Exercise 19.)

Further examples of power series that converge in the whole complex plane are given by the standard **trigonometric functions**; these are defined by

$$\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}, \quad \text{and} \quad \sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!},$$

and they agree with the usual cosine and sine of a real argument whenever $z \in \mathbb{R}$. A simple calculation exhibits a connection between these two functions and the complex exponential, namely,

$$\cos z = \frac{e^{iz} + e^{-iz}}{2} \quad \text{and} \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

These are called the **Euler formulas** for the cosine and sine functions.

Power series provide a very important class of analytic functions that are particularly simple to manipulate.

Theorem 2.6 *The power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ defines a holomorphic function in its disc of convergence. The derivative of f is also a power series obtained by differentiating term by term the series for f , that is,*

$$f'(z) = \sum_{n=0}^{\infty} n a_n z^{n-1}.$$

Moreover, f' has the same radius of convergence as f .

Proof. The assertion about the radius of convergence of f' follows from Hadamard's formula. Indeed, $\lim_{n \rightarrow \infty} n^{1/n} = 1$, and therefore

$$\limsup |a_n|^{1/n} = \limsup |n a_n|^{1/n},$$

so that $\sum a_n z^n$ and $\sum n a_n z^n$ have the same radius of convergence, and hence so do $\sum a_n z^n$ and $\sum n a_n z^{n-1}$.

To prove the first assertion, we must show that the series

$$g(z) = \sum_{n=0}^{\infty} n a_n z^{n-1}$$

gives the derivative of f . For that, let R denote the radius of convergence of f , and suppose $|z_0| < r < R$. Write

$$f(z) = S_N(z) + E_N(z),$$

where

$$S_N(z) = \sum_{n=0}^N a_n z^n \quad \text{and} \quad E_N(z) = \sum_{n=N+1}^{\infty} a_n z^n.$$

Then, if h is chosen so that $|z_0 + h| < r$ we have

$$\begin{aligned} \frac{f(z_0 + h) - f(z_0)}{h} - g(z_0) &= \left(\frac{S_N(z_0 + h) - S_N(z_0)}{h} - S'_N(z_0) \right) \\ &\quad + (S'_N(z_0) - g(z_0)) + \left(\frac{E_N(z_0 + h) - E_N(z_0)}{h} \right). \end{aligned}$$

Since $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1})$, we see that

$$\left| \frac{E_N(z_0 + h) - E_N(z_0)}{h} \right| \leq \sum_{n=N+1}^{\infty} |a_n| \left| \frac{(z_0 + h)^n - z_0^n}{h} \right| \leq \sum_{n=N+1}^{\infty} |a_n| n r^{n-1},$$

where we have used the fact that $|z_0| < r$ and $|z_0 + h| < r$. The expression on the right is the tail end of a convergent series, since g converges absolutely on $|z| < R$. Therefore, given $\epsilon > 0$ we can find N_1 so that $N > N_1$ implies

$$\left| \frac{E_N(z_0 + h) - E_N(z_0)}{h} \right| < \epsilon.$$

Also, since $\lim_{N \rightarrow \infty} S'_N(z_0) = g(z_0)$, we can find N_2 so that $N > N_2$ implies

$$|S'_N(z_0) - g(z_0)| < \epsilon.$$

If we fix N so that both $N > N_1$ and $N > N_2$ hold, then we can find $\delta > 0$ so that $|h| < \delta$ implies

$$\left| \frac{S_N(z_0 + h) - S_N(z_0)}{h} - S'_N(z_0) \right| < \epsilon,$$

simply because the derivative of a polynomial is obtained by differentiating it term by term. Therefore,

$$\left| \frac{f(z_0 + h) - f(z_0)}{h} - g(z_0) \right| < 3\epsilon$$

whenever $|h| < \delta$, thereby concluding the proof of the theorem.

Successive applications of this theorem yield the following.

Corollary 2.7 *A power series is infinitely complex differentiable in its disc of convergence, and the higher derivatives are also power series obtained by termwise differentiation.*

We have so far dealt only with power series centered at the origin. More generally, a power series centered at $z_0 \in \mathbb{C}$ is an expression of the form

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n.$$

The disc of convergence of f is now centered at z_0 and its radius is still given by Hadamard's formula. In fact, if

$$g(z) = \sum_{n=0}^{\infty} a_n z^n,$$

then f is simply obtained by translating g , namely $f(z) = g(w)$ where $w = z - z_0$. As a consequence everything about g also holds for f after we make the appropriate translation. In particular, by the chain rule,

$$f'(z) = g'(w) = \sum_{n=0}^{\infty} n a_n (z - z_0)^{n-1}.$$

A function f defined on an open set Ω is said to be **analytic** (or have a **power series expansion**) at a point $z_0 \in \Omega$ if there exists a power series $\sum a_n(z - z_0)^n$ centered at z_0 , with positive radius of convergence, such that

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n \quad \text{for all } z \text{ in a neighborhood of } z_0.$$

If f has a power series expansion at every point in Ω , we say that f is **analytic on Ω** .

By Theorem 2.6, an analytic function on Ω is also holomorphic there. A deep theorem which we prove in the next chapter says that the converse is true: every holomorphic function is analytic. For that reason, we use the terms holomorphic and analytic interchangeably.

3 Integration along curves

In the definition of a curve, we distinguish between the one-dimensional geometric object in the plane (endowed with an orientation), and its